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Artificial neural network-based forecasting of SPI for extreme event analysis in the Upper São Francisco sub-basin

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ABSTRACT

Droughts and floods impact environmental and socioeconomic systems, requiring tools for effective monitoring and management. This study applied artificial neural networks (ANNs) with a spatial approach to forecast the Standardized Precipitation Index (SPI) and assess extreme events in the Upper São Francisco River sub-basin. Multilayer feedforward ANNs, optimized via the Levenberg-Marquardt algorithm, were used for annual SPI prediction. SPI data from 1998–2015, obtained from the TRMM 3B42 product across 169 grid points, supported model training. The ANN#15 model, using 15 years of input data, showed superior performance, particularly for long-term SPI. Forecasts exhibited high to very high correlations ($R = 0.70$ – 1.00), with kappa values indicating excellent agreement ($\kappa = 0.81$ – 0.99). Some locations showed lower performance $\kappa = 0.20$ – 0.40 ; $R \approx 0.00$). Results confirm the model's ability to forecast SPI up to one year ahead, supporting its application in water resource management and climate adaptation.

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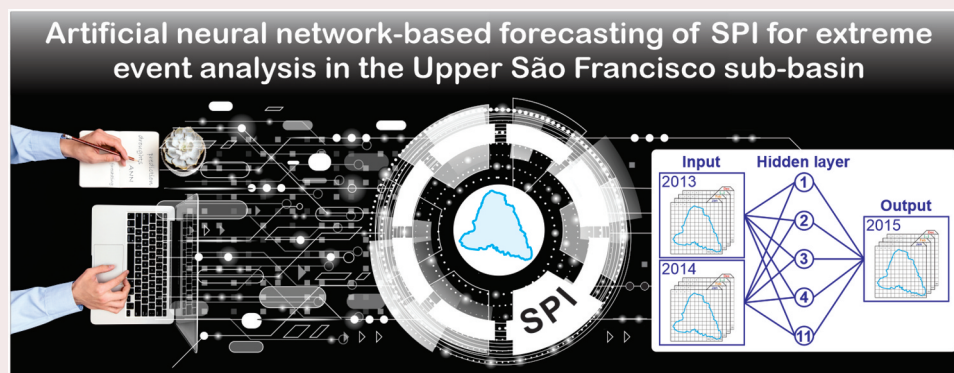
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1 Introduction

The occurrences of droughts and floods, inherently complex natural phenomena, have significant repercussions for ecosystems and socioeconomic, cultural, and historical activities (Istomina *et al.* 2023, Gebrechorkos *et al.* 2025). These events are characterized by gradual development, prolonged duration, and diverse manifestations within the same region (Rahmani and Fattahi 2024). Notably, water scarcity has emerged as a pressing global challenge, affecting vast areas across multiple continents over the past three decades and threatening the livelihoods of billions of people (Mishra and Singh 2010, Vallejo Orti and Negussie 2019, Penny *et al.* 2023, Wongchuig *et al.* 2023).

Beyond monitoring and analysing precipitation patterns, effective planning is essential – its success being intrinsically linked to the accuracy of associated forecasts (Zhong *et al.* 2026). In this context, hydrological modelling emerges as

a critical tool for both forecasting and decision-making. Artificial neural networks (ANNs) have proven particularly effective for modelling complex, non-linear, and non-stationary time series, making them valuable for predicting natural hydrometeorological events. Applications of ANNs span various domains, including rainfall-runoff modelling, erosion processes, and water quality forecasting (Mishra and Desai 2006, Cutore *et al.* 2009, Santos *et al.* 2009, 2019, Carneiro and Farias 2013, Farias and Santos 2014, Santos and Silva 2014, Le *et al.* 2015, Zerouali *et al.* 2023).

The impacts of hydrological extremes are directly influenced by the intensity and duration of dry and wet events. To adequately characterize and model such phenomena, hydrometeorological indices are widely employed, with the Standardized Precipitation Index (SPI) (McKee *et al.* 1993) being one of the most commonly used (Suliman *et al.* 2024,

Sunusi and Auliana 2025). The SPI has gained global recognition for its ability to quantify the intensity and duration of extreme precipitation events across various temporal scales and geographic contexts (Kamruzzaman *et al.* 2022, Brasil Neto *et al.* 2024). It provides a standardized and objective measure of drought or wetness severity based solely on precipitation anomalies and is extensively applied in meteorological and hydrological assessments (Santos *et al.* 2024, Venturi *et al.* 2025). Among its advantages, the SPI is computationally simple and enables spatial and temporal comparisons across different climatic zones (Tigkas *et al.* 2019). The use of satellite-derived precipitation data for SPI calculation has become increasingly common in hydrological studies, as orbital sensors provide rapid, wide-area coverage with continuously improving data quality (Gomaa *et al.* 2023). In Brazil, the SPI is officially recommended by national agencies, such as the National Institute of Meteorology (INMET), for identifying and monitoring precipitation anomalies (Soares *et al.* 2016, Brasil Neto *et al.* 2021).

Studies on the SPI are essential for identifying, monitoring, and evaluating climate variability associated with droughts and excess precipitation events (Fan *et al.* 2014). Because it allows analyses at multiple temporal scales, SPI supports the understanding of hydrological and agricultural impacts, informs water resource planning, and contributes to climate risk management and adaptation strategies. This is particularly relevant for large and strategic river basins with a key role in hydrological regulation, such as the Upper São Francisco River Basin (Koch *et al.* 2018). This basin plays a crucial role in hydrological regulation across a large portion of Brazil, supporting water availability in both humid and semi-arid regions. Its reservoirs and river network contribute to flow regulation, flood control, and drought mitigation, ensuring water security for human consumption, irrigation, hydropower generation, and ecosystems. In addition, the basin functions as a strategic hydrological corridor that integrates climatic variability and enhances regional resilience to extreme hydrological events.

The Upper São Francisco River Basin is also critical to Brazil's water and energy security, as it encompasses the Três Marias Hydroelectric Plant, one of the country's most important reservoirs for power generation and flow regulation (Páez Mendieta *et al.* 2024). Furthermore, the basin supports extensive agricultural production, urban water supply, and essential ecosystem services, making it vital to national socioeconomic and environmental stability (Vasco *et al.* 2024). Consequently, studying this basin is crucial not only for regional drought and flood management but also for ensuring water sustainability, food security, and energy resilience at the national scale (Santos *et al.* 2018).

Several previous studies have investigated drought behaviour and trends in the São Francisco River Basin. Santos *et al.* (2009) analysed three homogeneous precipitation zones between 1938 and 2005 using global wavelet spectra, SPI, and ANNs, demonstrating the potential of ANN-based SPI forecasting; however, their analysis was limited to three hydrological zones and did not provide a spatially distributed forecast across the entire basin. Paredes-Trejo *et al.* (2021) assessed drought conditions using satellite-based and ground-based indices, focusing on drought characterization rather than

forecasting. Their study evaluated extreme drought events in terms of occurrence, persistence, spatial extent, severity, and impacts on streamflow and soil moisture between 1980 and 2020, but did not address predictive modelling. Magalhães *et al.* (2025) investigated the propagation of meteorological droughts into hydrological droughts using probabilistic approaches, examining thresholds conditioned on drought duration and severity; however, their analysis did not involve SPI forecasting or machine learning techniques.

In contrast to these previous studies, which focused on a limited number of hydrological zones, specific drought indices, or isolated sub-basins, the present study provides a spatially explicit, basin-wide forecasting approach. Notably, this work represents the first application of ANN-based SPI forecasting implemented simultaneously across 169 TRMM-derived grid points, delivering a fully spatially distributed prediction framework for the Upper São Francisco River Basin. Unlike conventional approaches based on single time series or sparse raingauge networks, this methodology captures fine-scale spatial variability while maintaining an integrated representation of basin-scale hydroclimatic dynamics. Such an approach significantly advances drought and wetness monitoring and offers strong potential for operational water resources management and early-warning systems.

Although this study does not propose a new hydrological theory, its methodological innovation lies in scaling ANN-based forecasting from isolated time series to a spatial grid framework. This has direct practical implications for large-scale water resource management and drought adaptation strategies. Moreover, by producing one-year-ahead SPI forecasts across the entire sub-basin, the study provides factual and actionable information for drought and flood preparedness, reservoir operation, and climate adaptation in a region of critical social, economic, and energy importance to Brazil. Thus, the manuscript offers both a methodological contribution – through its spatial forecasting framework – and an applied contribution, by delivering results directly relevant to decision-making in water security.

Regarding the study period and data relevance, the TRMM 3B42 product remains a benchmark satellite precipitation dataset for global hydrology due to its consistent spatial and temporal coverage (Freitas *et al.* 2020). The use of this dataset ensured robust calibration and validation of the ANN models. Importantly, the proposed methodological framework is flexible and can be readily applied to more recent precipitation products, such as the Global Precipitation Measurement (GPM) mission (Huffman *et al.* 2020), ensuring that the insights obtained remain relevant for present and future applications.

Overall, while previous studies have employed ANNs or drought indices in the São Francisco River Basin, the present work complements and extends this literature by demonstrating how ANN-based forecasting can be scaled to simultaneous, spatially distributed time series, thereby enhancing operational drought monitoring strategies. Accordingly, this study applies ANN models within a spatially distributed framework to forecast SPI and analyse extreme precipitation events across the Upper São Francisco River sub-basin.

2 Materials and methods

2.1 Study area

The study area is located within the São Francisco Hydrographic Region, between latitudes 18°10' and 21°25'S and longitudes 43°75' and 46°75'W (Fig. 1). This sub-basin includes 106 municipalities and covers an approximate drainage area of 51 000 km². The Upper São Francisco River sub-basin is of critical importance due to its substantial water volume, which contributes to both energy generation and flow regulation within the basin. Notably, it hosts the Três Marias Hydroelectric Plant – one of Brazil's most significant reservoirs – granting the region considerable social, economic, and cultural relevance (Santos *et al.* 2017, Dantas *et al.* 2018).

2.2 Precipitation data

The data used in this study are derived from a spatial grid consisting of 169 equidistant points, each covering an area of 0.25° × 0.25°. This grid is located between latitudes 18° and 21°S and longitudes 43.75° and 46.75°W. Covering the period from 1 January 1998 to 31 December 2015, precipitation data were obtained from the 3B42 product of the Tropical Rainfall Measuring Mission (TRMM) satellite. These data are freely available at <https://pmm.nasa.gov/TRMM> (NASA – National Aeronautics and Space Administration n.d.). The data cover an area of approximately 130 000 km², encompassing the entire Upper São Francisco River sub-basin. Figure 1b shows the TRMM point grid used in this study.

2.3 Standardized Precipitation Index

In this study, the SPI, introduced by McKee *et al.* (1993), was used. This index evaluates precipitation deficits or surpluses across various temporal scales, effectively quantifying the impacts of droughts and floods in regions with differing water availability (McKee *et al.* 1995). SPI values can be calculated over multiple time scales, encompassing both short- and long-term conditions. Typically, scales of 1, 3, 6, and 9 months represent short-term assessments, whereas scales of 12, 24, and 48 months are used for long-term evaluations. Fundamentally, the SPI for a given time scale compares accumulated precipitation over that period in a particular year with precipitation totals for the same period across all years in the historical record. In this study, eight SPI time scales (1, 3, 6, 9, 12, 18, 24, and 48 months) were employed. The incorporation of multiple time scales allows for a more comprehensive assessment of both short- and long-term drought and wet conditions.

Events are categorized as dry when SPI values are negative and as wet when values are positive. The SPI classification system defines various categories arbitrarily split into (McKee *et al.* 1993): extremely dry (SPI < -2), severely dry (-2 ≤ SPI < -1.5), moderately dry (-1.5 ≤ SPI < -1), mildly dry (-1 ≤ SPI ≤ 0), mildly wet (0 < SPI ≤ 1), moderately wet (1 < SPI ≤ 1.5), severely wet (1.5 < SPI ≤ 2), and extremely wet (SPI > 2). The associated probabilities for SPI classification are based on the standard normal distribution curve. Due to its simplicity and robustness, and the ability to standardize across different climates and time scales, SPI has been widely adopted

for drought characterization and water resource management (Da Silva *et al.* 2024, Guria *et al.* 2025). This broad applicability makes SPI an indispensable tool in hydrological studies, aiding significantly in the management and planning of water resources in response to climatic variations.

2.4 Artificial neural network model

In this study, ANNs of the multilayer perceptron (MLP) type were employed, consisting of an input layer, a hidden (intermediate) layer, and an output layer. This architecture was selected due to its widespread adoption in the literature and its demonstrated effectiveness in forecasting and simulating water resource-related processes (Abbaszadeh *et al.* 2023, Truneh *et al.* 2024, Farias *et al.* 2025). Figure 2 illustrates the ANN architecture used in this study. Inspired by the functioning of the human brain, ANNs comprise interconnected artificial neurons. Their performance depends on a well-defined architecture – including network topology and activation functions – and an effective learning process, which involves adjusting synaptic weights (Freire and Santos 2020). Further details are available in ASCE Task Committee on Application of Artificial Neural Networks in Hydrology (2000a, 2000b) and Haykin (2011).

In this study, a total of 17 ANN models (ANN#1 to ANN#17) were developed and evaluated. The models differed in the length of the input datasets, which ranged from 1 to 17 years, and were tested across all eight SPI time scales. This multi-model approach enabled a comprehensive assessment of model performance and enhanced the robustness and reliability of the resulting conclusions. To predict the relationship between input and output data, each input variable $x_1, x_2, \dots, x_n$ is associated with a synaptic weight $w_1, w_2, \dots, w_n$, reflecting the relative importance of each input. The processing performed by the neuron's cell body is represented by the weighted sum of the inputs and their corresponding weights ($x_1w_1, x_2w_2, \dots, x_nw_n$). The activation function then receives this stimulus and transfers the neuron's response to the adjacent layer, playing a crucial role in determining the output values of the neurons ($y_1, y_2, \dots, y_k$) (Ghozat *et al.* 2023).

According to Haykin (2011), the sigmoid function is effective as an activation function and was used between the input and intermediate layers, while a linear function was applied between the intermediate and output layers. During the training process, algorithms adjust the synaptic weights to minimize the error between the produced output values and the desired ones at each neuron in the output layer.

Among the various training algorithms, back-propagation (Rumelhart *et al.* 2019), based on the gradient descent method, is the most widely used due to its ability to adjust weights considering the magnitude of the error. A common variation is the Levenberg-Marquardt algorithm, an extension of the Newton method, known for its rapid and efficient weight adjustment in ANNs. This algorithm was chosen for training the ANNs in this study.

In this study, 17 ANN models were constructed to forecast annual SPI values at multiple time scales (SPI-1, SPI-3, SPI-6, SPI-9, SPI-12, SPI-18, SPI-24, and SPI-48). The input datasets for these models ranged from one to 17 years. Consequently, each model, designated as ANN#1, ANN#2, ANN#3, ...,

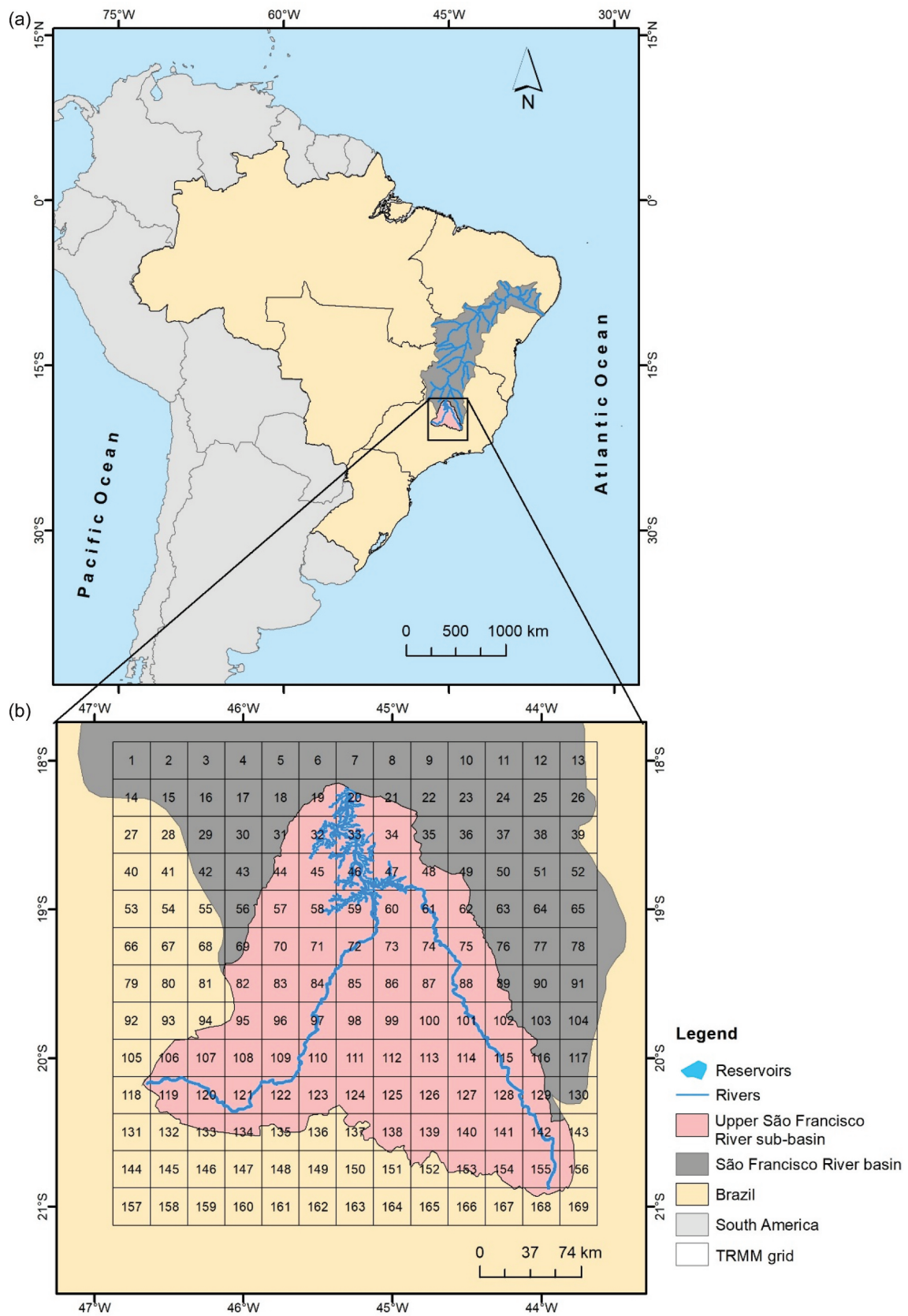


Figure 1. (a) Location of the Upper São Francisco River sub-basin in Brazil, and (b) Tropical Rainfall Measuring Mission (TRMM) grids in the study area.

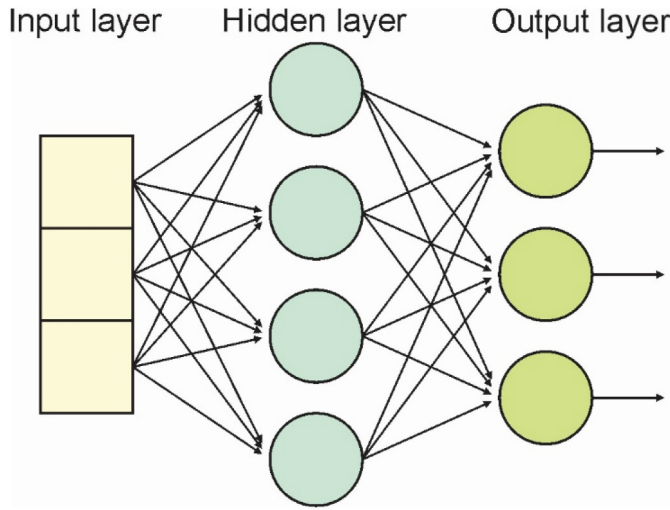


Figure 2. Structure of a multilayer perceptron (MLP) network with one intermediate layer.

ANN#17, employed a specific number of input years for SPI forecasting. For example, ANN#1 used input data from 2014 (one year) for all eight SPI time scales, whereas ANN#2 used data from 2013 and 2014 (two years) for the same scales. Accordingly, the dataset spanning from 1998 to 2014 (17 years) constituted the input data for ANN#17.

The number of neurons in the hidden layer of the ANN models was determined through systematic experimentation, following the approach proposed by Adil *et al.* (2022). For each model, different numbers of neurons – ranging from 1 to 20 – were tested. This process continued until the mean squared error (MSE) of the training set showed no substantial improvement. Based on these results, the optimal number of neurons for achieving the best statistical performance was identified as 11. This configuration was adopted as the standard for all ANN models in the study.

Figure 3 illustrates an example of ANN#2, aimed at predicting the values for the year 2015 based on SPI-1 input data from 2013 and 2014. The neural network is of the multilayer feedforward type, consisting of an input layer, an intermediate layer containing 11 neurons, and an output layer. As previously noted, sigmoid transfer functions are used between

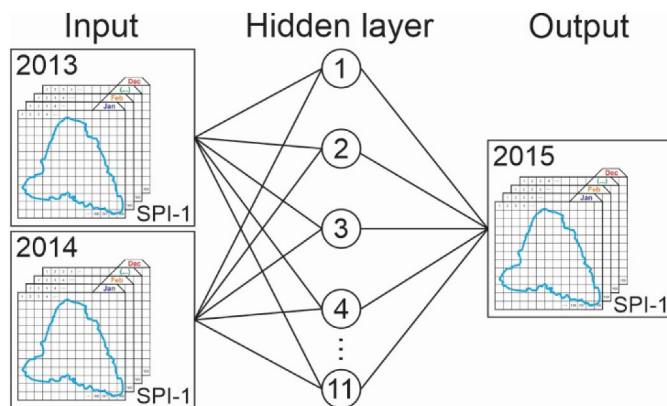


Figure 3. Structure of ANN#2 for annual forecasting of SPI-1.

the input and intermediate layers, while a linear function is employed between the intermediate and output layers.

The organization of SPI data for forecasting is illustrated in Fig. 4, using ANN#1 as an example. For this model, the input consists of SPI-1 values for all 12 months of the year 2014, spatially distributed over a 13×13 grid (169 points). Each month is represented by a two-dimensional matrix corresponding to the spatial configuration of the Upper São Francisco River sub-basin. These monthly matrices are then rearranged into one-dimensional sequences, preserving the spatial order of the grid points, to form time series inputs for the ANN. Based on these input sequences, the network predicts the corresponding SPI-1 values for the subsequent year (2015), generating 12 forecasted 13×13 matrices – one for each month – containing predicted SPI values for all 169 grid points. This procedure illustrates how spatially distributed SPI fields are transformed into ANN-compatible time series and subsequently reconstructed into spatial forecasts.

To perform the forecasting, the input data were divided into three sets: training, validation, and testing. These sets were partitioned sequentially, with 70% of the data allocated for training, 15% for validation, and 15% for testing. During training, the neural network was adjusted based on the prediction errors, and the training process was halted when generalization ceased to improve, as indicated by an increase in the MSE of the validation dataset. The validation data were used to evaluate the network's generalization capability. In contrast, the test data did not influence the training process and provided an independent assessment of the network's performance after training.

2.5 Performance evaluation

In this study, the performance of the ANN forecasts was evaluated using three statistical indices: the Pearson correlation coefficient (R), the coefficient of determination (R^2), and Cohen's kappa coefficient (κ). Higher values of R , R^2 , and κ , approaching unity, indicate better agreement between observed and predicted SPI values. The R and κ coefficients were computed as follows:

$$R = \frac{N \sum (SPI_i \widehat{SPI}_i) - \sum SPI_i \sum \widehat{SPI}_i}{\sqrt{[N \sum SPI_i^2 - (\sum SPI_i)^2] [N \sum \widehat{SPI}_i^2 - (\sum \widehat{SPI}_i)^2]}} \quad (1)$$

$$K = 1 - \frac{1 - p_o}{1 - p_e} \quad (2)$$

where SPI_i is the observed SPI at time scale i , $\widehat{SPI}_i$ is the corresponding forecasted value, p_o is the observed agreement, and p_e is the expected agreement due to chance.

The coefficient R measures the strength of the linear relationship between observed and predicted SPI values, with values close to 1 indicating a strong positive correlation. R^2 represents the proportion of variance in the observed SPI that is explained by the model predictions, providing an indication of predictive skill. Cohen's kappa

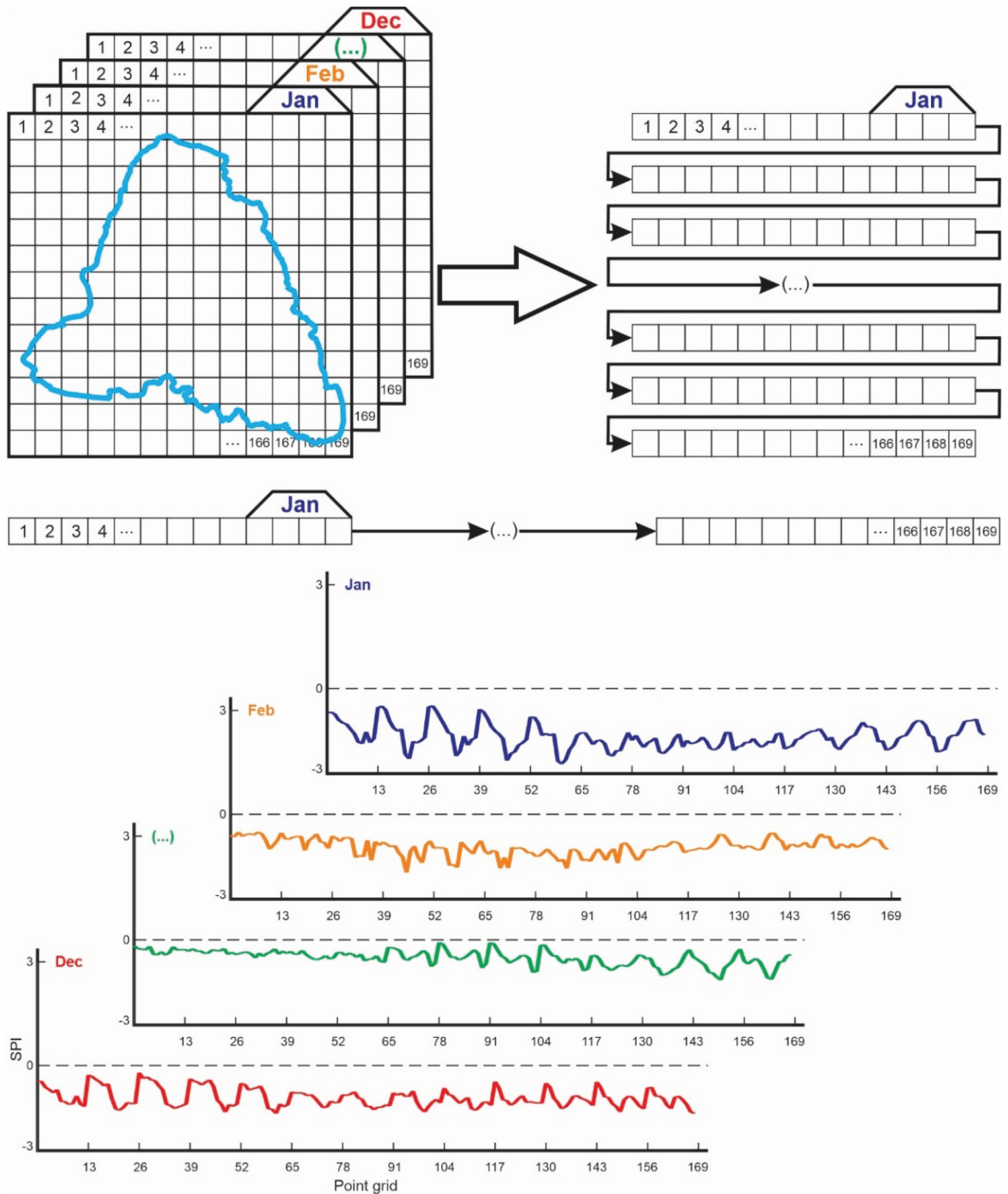


Figure 4. Representation of the data organization for conducting forecasts.

(κ), originally developed to assess agreement between categorical classifications, was adapted here to quantify the level of agreement between observed and predicted SPI classes beyond random chance. Values of κ close to 1 indicate strong agreement.

To ensure consistent interpretation of model performance, the values of R , R^2 , and κ were classified according to standard thresholds widely adopted in the literature. The classification criteria used in this study are summarized in Table 1, adapted from Hopkins (2002) for R and R^2 and from Landis and Koch

Table 1. Classification criteria for performance metrics used in model evaluation.

Performance level	Pearson correlation (R)	Coefficient of determination (R^2)	Cohen's kappa (κ)	Interpretation
Very low/poor	0.00–0.10	< 0.01	< 0.00	Very weak or no agreement
Low	0.10–0.30	0.01–0.09	0.00–0.20	Weak agreement
Moderate	0.30–0.50	0.09–0.25	0.21–0.40	Fair agreement
High	0.50–0.70	0.25–0.49	0.41–0.60	Moderate agreement
Very high	0.70–0.90	0.49–0.81	0.61–0.80	Strong agreement
Excellent	0.90–1.00	0.81–1.00	0.81–1.00	Almost perfect agreement

Source: Adapted from Hopkins (2002) for R and R^2 , and Landis and Koch (1977) for κ .

(1977) for κ . This unified classification framework facilitates transparent comparison of model performance across different SPI time scales and spatial locations.

3 Results and discussion

After evaluating 17 ANN models, the ANN#15 model, which used 15 years of input data (2000–2014), was identified as the best-performing model among those assessed. The scatter plot analysis of R and κ indicated that this model achieved more consistent and higher agreement values, highlighting its superior predictive performance compared to the other tested models. Figure 5 presents the scatter plot of R and κ values derived from all forecasts (12 months $\times$ 8 SPI scales) produced by ANN#15. The figure was generated by plotting the R and κ values obtained at each of the 169 grid points across the Upper São Francisco sub-basin. Each point in the scatter plot represents a single grid cell, resulting in a total of 169 plotted points.

Analysis of Fig. 5 reveals two instances in which the R coefficient was close to zero, indicating a very weak linear relationship, while the κ coefficient exceeded 0.70 in both cases. In another instance, although the κ value indicated weak agreement, the corresponding R value was high (0.70–0.90), resulting

in low dispersion around the line of equality. Despite these specific cases, the majority of the results were classified as having high to very high correlations and showed excellent agreement, based on the thresholds defined in Table 1, as evidenced by the overall distribution shown in Fig. 5.

Tables 2 and 3 display the monthly values of the R and κ coefficients resulting from forecasts made by the most effective ANN (ANN#15). The input datasets cover the period from 2000 to 2014, encompassing various SPI scales, while the output data consist of the forecast of the eight SPI scales for the subsequent year (2015). From Table 2, it is noted that the R values for long-term SPIs were better compared to the results for short-term SPIs. Among the short-term SPIs, SPI-9 showed low correlation between observed and predicted data for August and February 2015. Conversely, SPI-6 exhibited excellent correlation values, with R reaching 0.978 for March 2015. For long-term SPIs, all displayed good correlation coefficients, particularly SPI-24 in February and December, with the highest R values (0.982).

When analysing the kappa agreement degrees, as shown in Table 3, it is evident that the performance of the forecasts generally improves as the time scales of the SPI increase. There is relatively lower efficacy in the forecasts for short-

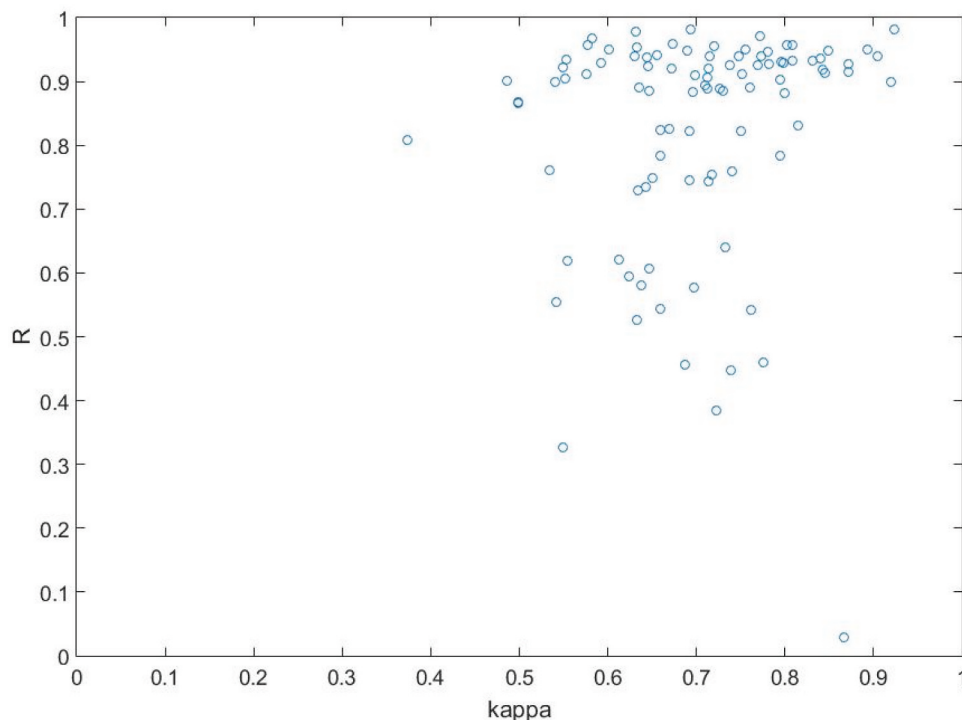


Figure 5. Scatter plot of the R and κ values resulting from the forecast at all 169 grid points using ANN#15.

Table 2. Correlation coefficients R between observed data and calculations by ANN#15 for the months of 2015 and SPIs at different temporal scales.

Months	SPI-1	SPI-3	SPI-6	SPI-9	SPI-12	SPI-18	SPI-24	SPI-48
January	0.326	0.938	0.929	0.448	0.932	0.899	0.950	0.971
February	0.578	0.62	0.956	0.028	0.891	0.927	0.982	0.959
March	0.922	0.607	0.978	0.784	0.940	0.884	0.948	0.939
April	0.748	0.754	0.967	0.744	0.925	0.831	0.929	0.958
May	0.743	0.92	0.934	0.823	0.910	0.823	0.933	0.950
June	0.759	0.594	0.867	0.459	0.906	0.940	0.931	0.942
July	0.555	0.808	0.899	0.542	0.885	0.882	0.939	0.953
August	0.621	0.729	0.911	-0.014	0.895	0.888	0.926	0.924
September	0.457	0.64	0.868	0.527	0.928	0.957	0.939	0.950
October	0.385	0.783	0.891	0.734	0.919	0.921	0.936	0.956
November	0.544	0.886	0.904	0.822	0.911	0.946	0.904	0.948
December	0.761	0.825	0.901	0.581	0.915	0.914	0.982	0.889

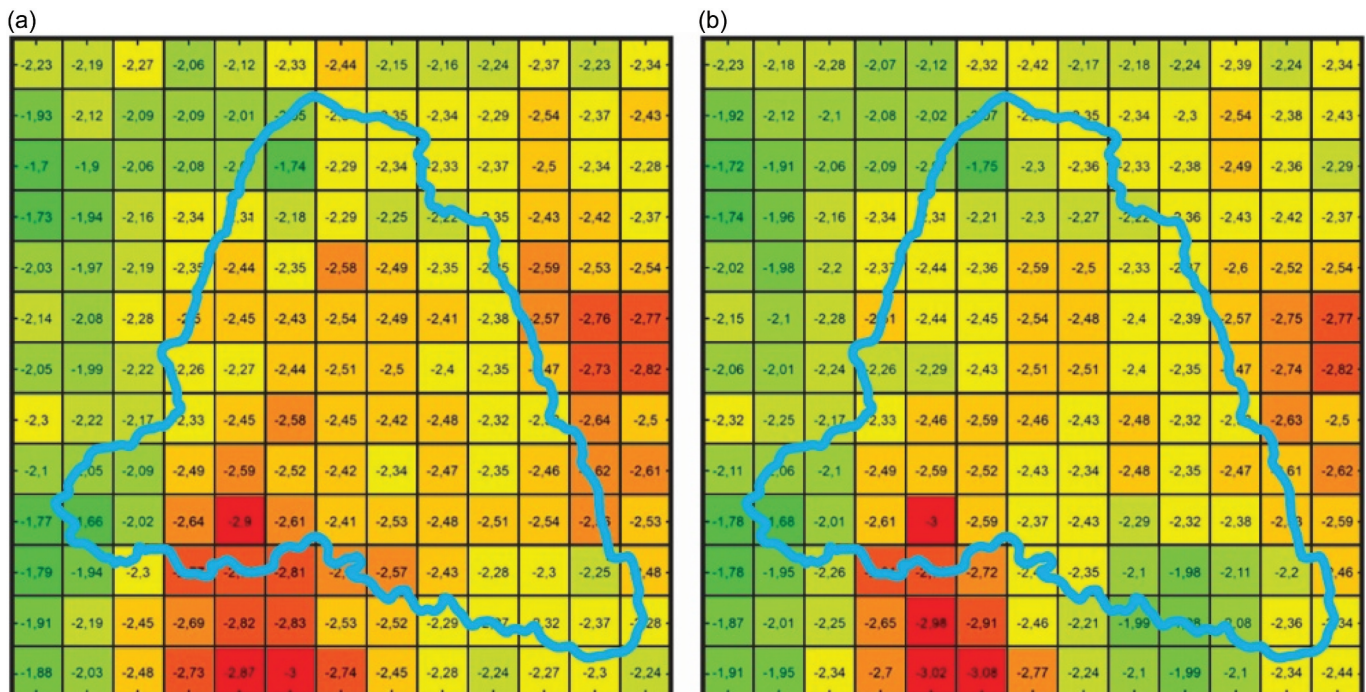
Table 3. Kappa concordance index (κ) between observed data and calculations by ANN#15 for the months of 2015 and SPIs at different temporal scales.

Months	SPI-1	SPI-3	SPI-6	SPI-9	SPI-12	SPI-18	SPI-24	SPI-48
January	0.549	0.644	0.592	0.740	0.831	0.920	0.894	0.772
February	0.697	0.555	0.809	0.866	0.761	0.872	0.924	0.673
March	0.549	0.647	0.632	0.795	0.904	0.696	0.849	0.715
April	0.650	0.717	0.583	0.692	0.769	0.815	0.798	0.577
May	0.714	0.714	0.553	0.751	0.698	0.659	0.808	0.756
June	0.740	0.623	0.499	0.775	0.712	0.630	0.796	0.655
July	0.542	0.374	0.541	0.762	0.730	0.800	0.747	0.632
August	0.613	0.635	0.576	0.727	0.710	0.712	0.738	0.645
September	0.687	0.732	0.499	0.633	0.782	0.803	0.774	0.602
October	0.722	0.660	0.635	0.643	0.842	0.672	0.840	0.720
November	0.659	0.646	0.552	0.693	0.752	0.780	0.794	0.690
December	0.534	0.669	0.486	0.872	0.872	0.846	0.694	0.727

term SPIs, while the concordance values are higher for long-term SPIs. This trend can be explained by the greater temporal variability of precipitation in short-term SPIs, while such variability is mitigated at longer time scales due to the inclusion of more monthly data.

To provide a clearer and more illustrative representation of the results presented in Tables 2 and 3, Fig. 6 displays the

spatial distribution of observed and predicted SPI-24 values for January 2015. The results indicate that the ANN#15 model successfully detected both extreme dry and wet events, particularly at longer SPI time scales (e.g. SPI-24). For example, in January 2015, the model accurately reproduced extremely dry conditions ($\text{SPI} < -2.0$) observed at specific grid points within the sub-basin, while also capturing wet anomalies ($\text{SPI} > 2.0$)

**Figure 6.** Comparison of (a) observed SPI-24 values and (b) SPI-24 values predicted by ANN#15 for January 2015 in the Upper São Francisco River sub-basin.

in other localized areas. This ability to identify extremes demonstrates the robustness of the model in forecasting critical hydrological events, which is particularly relevant for water resource management and drought preparedness. Moreover, the strong performance of ANN#15, reflected by high correlation and kappa values at long-term SPI scales, supports its application in reservoir management, agricultural planning, and drought preparedness. Particular attention is given to the operational importance of forecasting SPI one year ahead for the Três Marias Reservoir, which plays a critical role in national hydropower generation and water allocation.

In Fig. 6, a gradient colour scheme (red, orange, yellow, light green, dark green) is also used to denote the variation of SPI-24 values, with the lowest values encoded in red, progressing through orange, yellow, and light green, culminating in the highest values represented in dark green. Reviewing Fig. 6, it is observed that for January, the forecast was highly satisfactory, as all 169 predicted points displayed values very close or identical to those observed. Moreover, the effectiveness of the prediction, evidenced by a κ value of 0.894, was categorized as excellent (0.81–0.99) in terms of agreement. The spatialized SPI-24 data (13×13 matrix) in Fig. 6 represents two time series, highlighting the observed SPI-24 series in January 2015 versus the forecasted for the same period in the year 2015.

Figure 7 illustrates the spatial distribution (13×13 matrix) of the observed and predicted SPI-24 values for January 2015 obtained with ANN#15, according to the adopted classification. As shown in Fig. 7, the values ranged from extremely dry (SPI < -2.00) to severely dry ($-2.00 \leq \text{SPI} < -1.50$), corresponding to classification bands 1 and 2, respectively. Moreover, the

deviation of κ from 1 can be attributed to discrepancies between the predicted and observed classifications at specific points, such as 80, 140, 152, 153, 158, and 166. These mismatches were localized and dispersed rather than organized into a well-defined continuous spatial cluster.

Figure 8 presents a comparison between the observed and predicted SPI-24 data for all months of 2015 across the region. The “Observed SPI-24” column displays the spatialization of observed values, while the “Predicted SPI-24” column shows the data predicted by ANN#15 using input SPI-24 data from 2000 to 2014. The graphs to the right of each month represent the scatter between observed and predicted data, comparing them against the line of equality and providing the values of the coefficients R and κ .

The analysis of the spatial distribution of drought indicates that from April to December, the eastern and northeastern parts of the study area were more severely affected, while the western portion experienced less impact. The central region was also characterized by a high severity of drought, reflected in extremely negative values of SPI-24. Conversely, in January and February, the southern and southwestern regions stood out as the most affected by drought, exhibiting the highest negative values of SPI-24.

Regarding the correlation between observed and predicted SPI-24 values obtained with ANN#15 for all months of 2015, the values of R ranged from 0.904 in November to 0.982 in February and December, while the values of κ varied from 0.694 in December to 0.924 in February. These results indicate consistently high correlations and a stable prediction quality throughout 2015 for SPI-24.

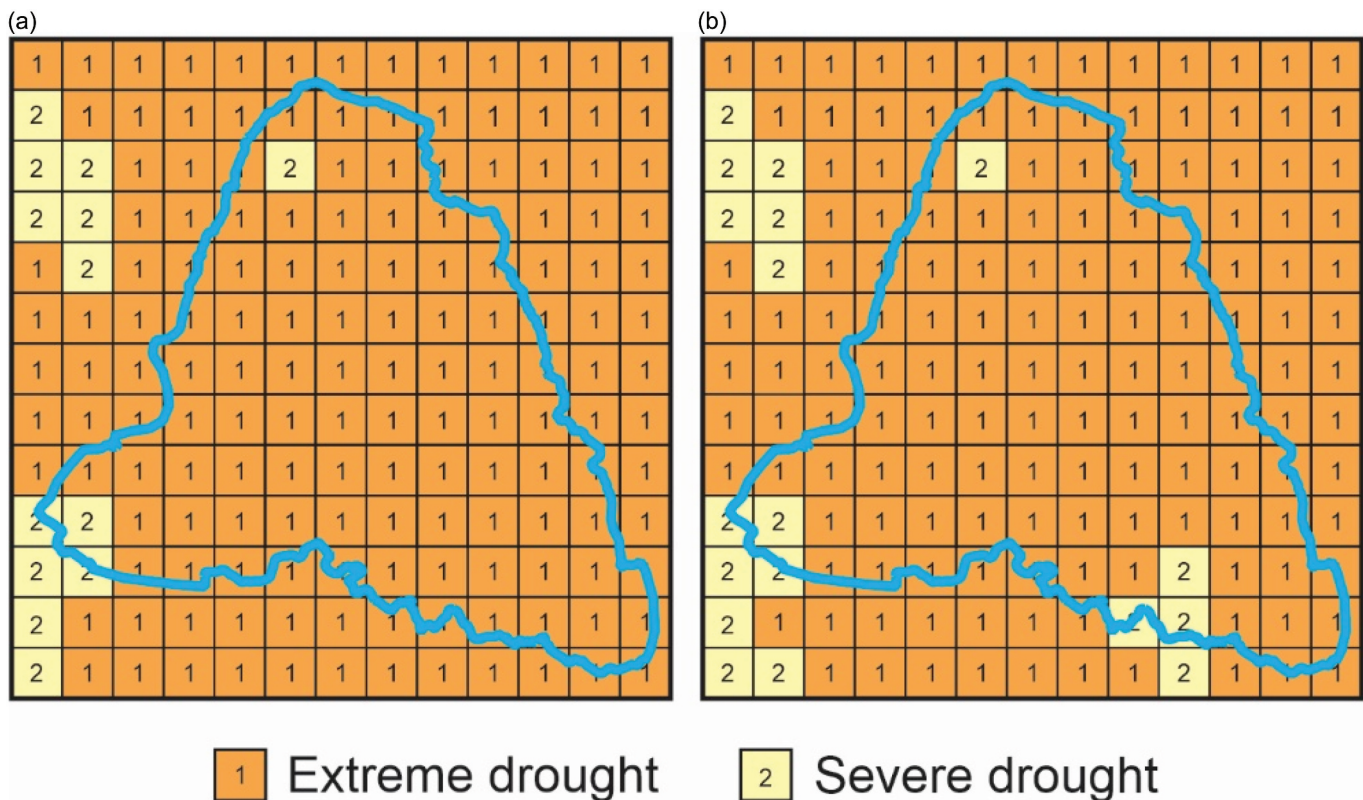


Figure 7. (a) Observed SPI-24 values and (b) SPI-24 values predicted by ANN#15, classified into two SPI categories for January 2015.

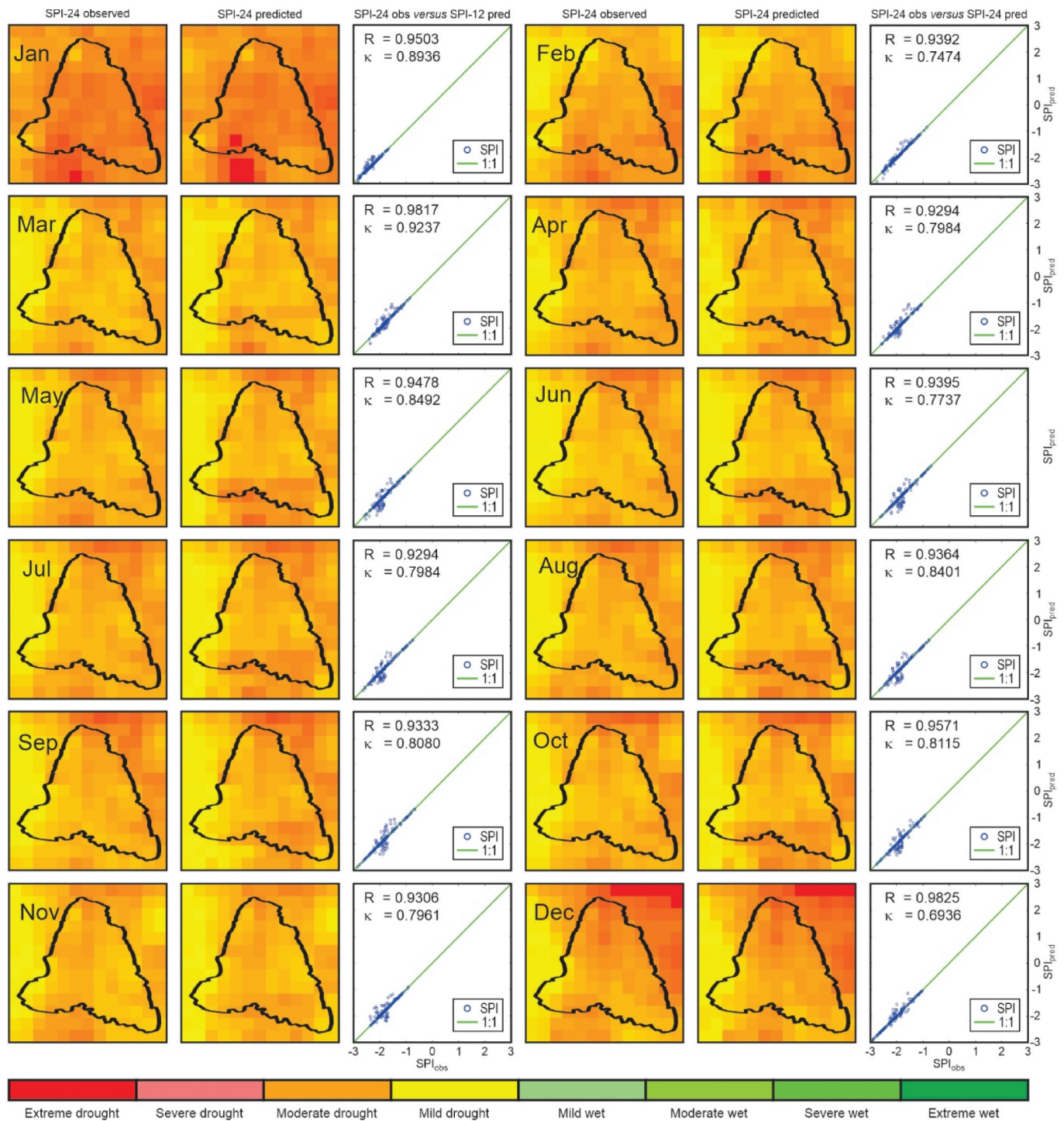


Figure 8. Comparison of observed and predicted SPI-24 values by ANN#15 for the year 2015 in the Upper São Francisco River sub-basin.

Nevertheless, some localized reductions in model skill were observed. Similar point-specific decreases in predictive performance have been reported in satellite-based hydroclimatic studies and are commonly associated with strong spatial variability in precipitation, limitations of remote sensing products, and non-linear hydrological responses that are not fully captured by predictive models (Huffman *et al.* 2007, Kidd and Levizzani 2011). In addition, the reliance on TRMM data from 1998 to 2015 may have influenced performance at specific grid points, particularly in areas dominated by convective rainfall

processes (Salas *et al.* 2000). Despite these localized limitations, the overall results demonstrate robust predictive performance across most of the basin, reinforcing the capacity of ANN models to represent complex hydroclimatic relationships (Maier *et al.* 2010, Tripathy and Mishra 2024).

This comprehensive analysis, supported by graphical representations and statistical metrics, underscores the efficacy of ANN#15 in providing accurate and reliable SPI forecasts across multiple temporal and spatial scales, demonstrating its potential to enhance water resource management and climate

change adaptation strategies in hydrologically sensitive regions. Moreover, the successful reproduction of both extreme dry and wet anomalies, particularly at longer SPI time scales, highlights the practical value of the proposed framework for drought preparedness, reservoir operation, and agricultural planning. This is especially relevant for the Três Marias Reservoir, which plays a critical role in national hydropower generation and water allocation.

Although this study focused on the Upper São Francisco River sub-basin, the application of ANN models using spatially distributed precipitation data is transferable to other river basins. By replacing the input precipitation dataset and recalibrating the models according to local hydroclimatic conditions, the proposed approach can be readily adapted to regions exposed to drought or flood risks. The methodological framework presented in this study therefore exhibits strong transferability and broad applicability across different river basins and climatic contexts.

Because the approach relies on gridded precipitation datasets rather than dense ground-based monitoring networks, it is particularly suitable for data-scarce regions where conventional hydrometeorological observations are limited. The integration of ANNs with spatially distributed precipitation inputs facilitates adaptation to basins with diverse hydroclimatic regimes, physiographic characteristics, and spatial scales. By recalibrating the models using locally relevant precipitation data, the framework can be applied to regions facing a wide range of climate-driven challenges, including recurrent droughts, extreme rainfall events, and increasing hydroclimatic variability. This flexibility extends the relevance of the proposed methodology beyond the case study basin, positioning it as a robust tool for comparative hydrological analyses, early-warning systems, and climate change adaptation strategies. Consequently, the framework contributes meaningfully to advances in basin-scale hydroclimatic forecasting under evolving climatic conditions.

4 Conclusion

This study aimed to develop and evaluate ANNs for space-dependent forecasting of the SPI in the Upper São Francisco River sub-basin. Among the 17 models tested, ANN#15, which utilized 15 years of input data (2000–2014), demonstrated the best predictive performance. Most forecasts achieved high (0.70–0.90) to very high (0.90–1.00) correlation coefficients, with κ values classified as excellent (0.81–0.99). Only a limited number of grid points exhibited weaker performance, characterized by very low correlations ($R \approx 0.00$) and fair agreement ($\kappa = 0.20$ –0.40). Despite these localized discrepancies, the overall results indicate robust predictive capability across most of the basin.

The results also showed that long-term SPI forecasts outperformed short-term forecasts. In most cases, predictive performance improved with increasing SPI time scales, reflecting the attenuation of short-term precipitation variability when longer accumulation periods are considered.

Overall, the findings confirm that ANNs are effective tools for space-dependent SPI forecasting up to one year in advance. The inclusion of the coefficient of determination (R^2) further strengthened model evaluation by quantifying the proportion of explained

variance in SPI forecasts. A key contribution of this study lies in the implementation of an ANN framework spatially distributed across 169 grid points, enabling basin-wide forecasting rather than prediction at isolated stations or a limited number of locations. This approach enhances the representation of localized drought and wet anomalies while maintaining an integrated basin-scale perspective. The proposed framework therefore shows strong potential to improve predictive water resource management, drought preparedness, reservoir operation, and climate adaptation strategies in hydrologically sensitive regions.

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Data availability statement

Data will be made available on reasonable request.

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